

Grade 8
Unit 8
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

1. Determine whether each value of k is a solution to the equation $2k + 1 = k + k + 1$.

Value of k	$k = 0$	$k = 0.5$	$k = 1$	$k = -1$	$k = 8$	$k = -3$
Solution?	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No

2. Based on your work in question #1, explain how many solutions there are to the equation $2k + 1 = k + k + 1$.

Infinitely many
-Explanations will vary.

3. Determine whether each value of m is a solution to the equation $8m + 1 = 2(4m + 1)$.

Value of m	$m = 0$	$m = 8$	$m = -1$	$m = 4$	$m = -0.5$	$m = -4$
Solution?	☐ Yes ☒ No	☐ Yes ☒ No	☐ Yes ☒ No	☐ Yes ☒ No	☐ Yes ☒ No	☐ Yes ☒ No

4. Based on your work in question #3, explain how many solutions there are to the equation $8m + 1 = 2(4m + 1)$.

No solution
-Explanations will vary.

5. Samir claims there are infinitely many solutions to the equation $4a + 1 = 4a + 1 + a$ because both sides of the equation have $4a + 1$. Explain whether you agree with Samir.

No, because the right side of $=$ can be simplified to $5a + 1$, therefore both sides are not the same.

For each equation, simplify each side of the equation by combining like terms or applying the distributive property in the first column. Then, use the rewritten form of the equation to determine if it has one solution, no solutions, or infinitely many solutions.

	Equation	One Solution	No Solutions	Infinitely Many Solutions
6.	$3f + 14 = 9 + 3f + 5$ $3f + 14 = 3f + 14$	O	O	☒
7.	$5(2h + 3) = 10h + 3$ $10h + 15 = 10h + 3$	O	☒	O
8.	$3b + b - 9 = 2b + 2b + 9$ $4b - 9 = 4b + 9$	O	☒	O
9.	$3(-4j + 1) = 5(-4j + 2)$ $-12j + 3 = -20j + 10$	☒	O	O
10.	$3y + 12 + y = 4(y + 3)$ $4y + 12 = 4y + 12$	O	O	☒